

Technical Document #192: Deep Learning - Comprehensive Overview

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Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Dropout is a regularization technique where randomly selected neurons are ignored during training. This prevents co-adaptation and improves generalization.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies.
- Recurrent neural networks (RNNs) are designed to process sequential data.
- Attention mechanisms allow models to focus on different parts of the input when producing output.